



Governing industrial estates on Jakarta's periurban area: From shadow government to network governance

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This article investigates the way in which periurban politics have mediated foreign direct investment relocations and facilitated the spatial restructuring of the Jakarta Metropolitan Area (JMA), one of the largest and fastest-growing megaurban regions in Asia. We conducted a series of in-depth interviews in Cikarang, the largest, most developed industrial estate corridor in the urban periphery of JMA. We identified institutional settings, power struggles and governance mechanisms underlying the industrial estate transitions in the past three decades. We found that periurban governance in JMA has resulted in poor infrastructure connectivity and tight interactor and inter-regional competition. As global economic turbulence and decentralization policies restrict the effectiveness of privatization strategies, network strategies are trialled to promote sustainability and inclusiveness in JMA's industrial estates and towns.

Keywords: FDI, governance, industrial estate, Jakarta Metropolitan Area, periurbanization, politics

Introduction

Globalization of investments has long contributed to Indonesia's economic growth and spatial development. During the Dutch colonial era in the nineteenth and twentieth centuries, large-scale foreign investment projects in agricultural, agri-industrial and mining sectors changed the traditional rural landscapes in Java, Sumatera and Kalimantan (Lindblad, 2008). Because of the industrialization policy in Suharto's era (1966–98), foreign direct investment (FDI) in manufacturing since the 1980s has driven the expansion of mega property development in the form of regional infrastructure and private industrial estates in the periurban areas of the Jakarta Metropolitan Area (JMA) (Firman, 1998).

Periurban areas, periurban fringes or periurban interfaces originally refer to the transitional zones between the urban and the rural. In most developing and transitional countries in Africa, Asia and Latin America, periurban areas consist of urbanizing areas that are functionally integrated and extend from existing cities but are physically located outside designated city boundaries (Simon, 2004; 2008). In their early development, periurban areas in Asia may take the form of 'desakota', which refer to zones of sporadic, gradual rural to urban transition, where agricultural, industrial and urban activities coexist and intermingle intensively (McGee, 1991). Periurban areas may occupy wide regions with radius of tens to hundreds kilometres away from the main city centres. Recently, they are often seen to be the most problematic part of an urban, metropolitan or megaurban region, because of their mixed and dynamic land uses, rapid transformation and globally induced industrial development (Webster, 2002; Aguilar *et al.*, 2003; Hudalah *et al.*, 2007; Keivani & Mattingly, 2007).

This article attempts to respond to the lack of knowledge on the political and institutional complexity of globally linked rapid periurbanization, i.e., urban development

processes in the outskirts of established city boundaries (Keivani & Mattingly, 2007). It approaches the complexity by drawing on the shifting conceptual paradigm between government and governance. First, the term 'government' highlights the use of monopolistic and coercive power (of the state) in maintaining public order, coordinating interests and enabling collective action (Stoker, 1998). Meanwhile, for the same outputs, the governance perspective emphasizes innovative processes, noncoercive mechanisms and informal institutional arrangements in the context of blurring boundaries between the state, market and society (Bevir, 2011). Using the institutionalist approach, Pierre (1999) argues that different modes of governance may be formed in different cities, depending on the systems of political, economic and social values and beliefs within which the cities are situated. The general criteria used in evaluating modes of governance include key participants and their relations, the political objectives and the main instruments employed to achieve the objectives (DiGaetano & Strom, 2003).

This article particularly seeks to explore the governance formation surrounding *kawasan industri*, or industrial estates, and town formation in JMA. With a total population of more than 22.5 million, JMA is Indonesia's largest periurbanizing region with a multitude of challenges, including but not limited to environmental degradation, social segregation and intergovernmental coordination (Firman, 2009; Hudalah & Firman, 2012). This article addresses several questions: To what extent have domestic institutional settings played a role in the realization of FDI relocations in periurban JMA? How are FDI relocations interlinked with the periurban governance? How can this global-local nexus explain the periurban spatial restructuring in JMA?

The main case study is the Cikarang industrial corridor, located in Bekasi Regency and about 35–40 km east of Jakarta. Covering a planning area of almost 15 000 ha, private industrial estates in Cikarang have become Indonesia's largest manufacturing powerhouse. A recent study shows that Bekasi Regency, with a total population of 2 989 198 (The Indonesia Central Bureau of Statistics, 2014), captures about 50 per cent of JMA's FDI in the secondary sectors, which is mostly invested in Cikarang (Hudalah & Firman, 2012). The same study also emphasizes that manufacturing contributes to almost 80 per cent of the gross domestic product (GDP) of this regency. These industrial estates accommodate more than 3000 companies from over 30 countries (The Indonesia Central Bureau of Statistics, 2005). They employ at least 532 814 local and foreign workers (West Java Provincial Government, 2010). Their employment structure is typified by high-technological and heavy industries such as automotive, electronics, chemical, machinery, paper, metals and plastics (Hudalah *et al.*, 2013).

This article presents a brief note on methodology, followed by the role of three main actors or factors contributing to the industrial estate development: the central government in creating a broader context of FDI and manufacturing relocations in JMA; private developers in periurban spatial restructuring in JMA since the 1980s; and domestic institutional settings in reshaping local government in the periurban area, notably the country's transition to democratic decentralization. The main analysis explores the dynamics of governance mechanisms underlying periurban settlement transitions in Cikarang in the past three decades. The article concludes that a 'shadow government'—referring to active, exclusive roles of private developers operating outside existing representative government structures—was a very influential mode of governance during the early formation of the industrial estates and towns. Furthermore, a new mode of governance has been trialled, stressing the importance of networking, cooperation and collaboration among developers and between the private and public sectors in addressing spatial fragmentation and poor infrastructure connectivity.

Methodology

The research uses a mixed-methodological approach. I collected secondary data from statistical reports, press releases, official documents and newspaper articles, to explain FDI relocations in JMA and spatial restructuring in the case study. Furthermore, to uncover the dynamics of periurban politics and governance, I conducted a series of informal interviews with those directly involved in the spatial and governance transformation in the Cikarang industrial corridor since the 1980s. The respondents included local and provincial government officials, private developers, industrial estate and town managers, the industrial developers' association, industrial town planners, urban designers and architects. The interview questions in this study mainly elicited respondents' background, views and experiences in the building of the global periurban area of Cikarang. I conducted 20 individual and group interviews and several follow-up interviews between 2012 and 2013. Each interview lasted for about one hour.

The main analysis of historical data in this study explores the dynamics of competition, conflicts, confrontations and cooperation among different participating groups with different interests at different spatial scales through time. The in-depth governance analysis pays particular attention to transport infrastructure networks, which are considered to be a defining yet neglected spatial element of periurban areas (Hudalah *et al.*, 2007).

Industrial estate development from 1980s to 2010s

The central government policy and surge and fall of FDI in JMA

The second half of the 1980s marked a worldwide expansion of FDI. According to Urata (1993), apart from the strong global economic environment at that time, this rapid FDI expansion was seen as a response of major newly developed countries such as South Korea, Taiwan, Hong Kong and Singapore to a declining competitiveness of their manufacturing products. In these countries, higher production costs associated with stricter environmental regulations, rising wages and land rent, and currency revaluation resulted in inefficient production systems (Wie, 1991). Furthermore, as the newly developed countries' economic structure stabilized, strong government intervention in the manufacturing sector increased the transaction costs and thus hindered innovation and expansion of companies in their home countries (Hatch & Yamamura, 1996). These combined production and institutional factors particularly forced Japanese, and later, Taiwanese and South Korean companies to rapidly relocate their manufacturing functions to emerging Southeast Asian countries, including Indonesia.

Since 1966, the New Order regime led by former President Suharto has quickly liberalized the Indonesian economy. To boost the national economic growth and improve the standard of living, the central government periodically designed market-oriented and outward-looking policies. It gradually simplified laws and regulations to create more favourable conditions for private participation and foreign investments. From the 1980s to 1990s, the government focused its economic policies on reducing dependence on oil and gas revenues, by capturing the upcoming global relocation of export-oriented industries mainly from Japan and the four newly industrialized countries of South Korea, Taiwan, Hong Kong and Singapore (Wie, 1991). During this period the government progressively extended the liberal and market-oriented policies through massive enactment of deregulation packages in trade, financial, industrial and property sectors (Henderson & Kuncoro, 1996; Firman, 1998; Tambunan, 2008).

Among other instruments enacted to help speed up the absorption of FDI in manufacturing, the Presidential Decree No. 53 Year 1989 provided a legal basis for private and foreign companies to actively participate in industrial land development and management (Government of the Republic of Indonesia, 1989). The decree also highlighted that manufacturing activities had to be planned and concentrated in industrial estates with the provisioning of infrastructure and supporting environmental facilities, and emphasized that industrial estates must be developed and operated by licensed companies.

Since the exchange rate devaluation and liberalization policies of the late 1980s, Indonesia has increasingly enjoyed an FDI boom period (Hill, 1988). Most large-scale, export-oriented FDI in manufacturing is concentrated in areas within a distance of 50 km from major cities or around the four largest metropolitan regions in Java, namely JMA, Surabaya, Bandung and Semarang (Henderson & Kuncoro, 1996; Kuncoro, 2000; Kuncoro & Wahyuni, 2009). The largest concentration of globally-linked manufacturing can be found in West Java, especially in JMA, Indonesia's largest urban and industrial agglomeration (Hill, 1990). Most FDI within JMA is increasingly located in the periurban areas mainly consisting of regencies and municipalities bordering with Jakarta: Bogor, Depok, Tangerang and Bekasi (also known as Bodetabek) (Figure 1). According to the data provided by the Indonesia Investment Coordinating Board (2014), in 1996, FDI in the periurban areas accounted for USD 12.5 trillion, while the metropolitan core absorbed only USD 1.2 million.

However, in 1997 uncontrolled flows of short-term transnational investment triggered the Asian economic crisis. The financial crisis badly hit Indonesia, causing the FDI in JMA to dramatically drop to the lowest level in five years. In this period, the periurban areas suffered the most, because the FDI was mostly concentrated in manufacturing and foot-loose industries, which are highly dependent on foreign inputs and have a limited technological lifetime (of 10–15 years). While other Asian countries quickly recovered, Indonesia descended into multidimensional economic and political crises. The national economy took a decade to recover; only in the second half of the 2000s did FDI start to grow again, first in the core of the JMA, and later in its periurban areas.

From 2007 to 2009, the integration of local and global financial markets led to the global financial and European debt crises. However, Indonesia was not significantly

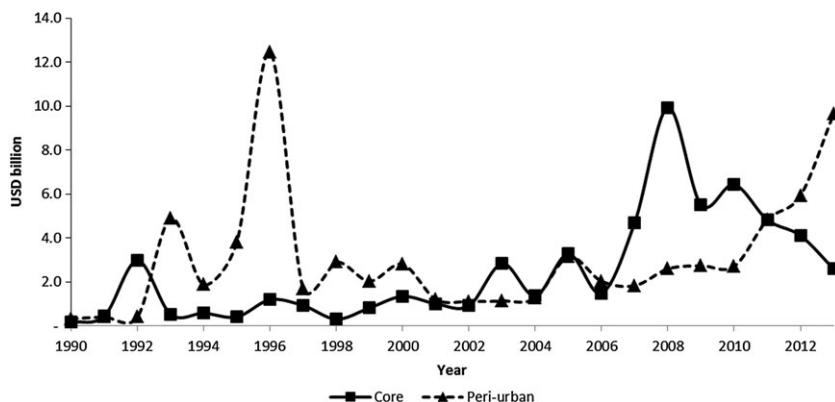


Figure 1. FDI in Jakarta Metropolitan Area from 1990 to 2012.

Source: Analysed from the Indonesia Investment Coordinating Board (2014).

affected, because of its stable political environment, stronger macroeconomic foundations and a large domestic market. In fact, together with China and India, two other large emerging Asian countries, Indonesia has become a bright spot for FDI in the face of the global uncertainty (CCTV English, 2012). As a result, since the beginning of the 2010s, FDI in periurban Jakarta has picked up, almost reaching the level of the 1990s boom period. The proportion of FDI in periurban Jakarta has taken over again the proportion of FDI in the metropolitan core.

The spatial restructuring of the periurban area of Jakarta by private developers

In addition to regulatory and policy reforms, major regional infrastructure projects initiated by the central government have contributed to the periurbanization of FDI in emerging Asian countries (Douglass, 2000; Wei & Leung, 2005). Likewise, the rapid ribbon-like spatial pattern of urban and industrial development in JMA cannot be separated from the construction of intercity toll roads in the 1980s, which connected Jakarta's downtown, seaport and airport with the neighbouring peripheral areas (Rimmer & Dick, 2009). Particularly, Jakarta-Merak Intercity Toll Road (98 km) and Jakarta-Cikampek Intercity Toll Road (83 km) built along the coastal zone are the physical foundation for the emergence of globally linked industrial estates and towns along the western and eastern periurban corridors, including Cikarang in Bekasi Regency (Figure 2).

In addition to regional accessibility, private industrial developers also consider the availability of large amounts of cheap and vacant land to locate new factories and housing facilities in the urban peripheries (Rimmer & Dick, 2009). Cikarang formerly consisted of arid land, excavated for the production of roof tiles and bricks. However, within a

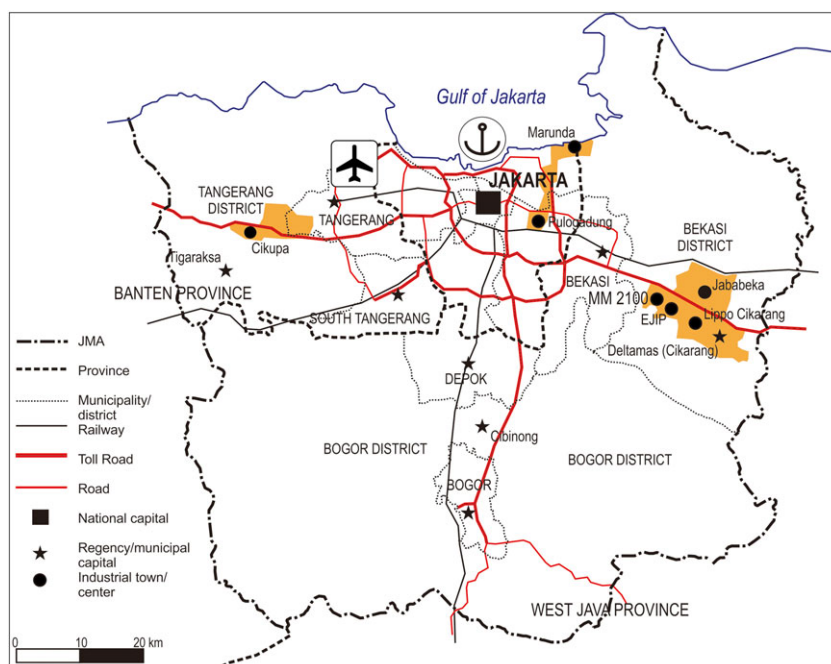


Figure 2. Map of Jakarta Metropolitan Area (JMA).

Source: Adapted and updated from Hudalah & Firman, 2012.

decade, the periurban area has dramatically transformed from this unfertile, neglected, rural hinterland into JMA’s largest industrialized and urbanizing periphery (Figure 3).

Seven private developers (industrial estate companies) can currently be found in Cikarang (Table 1). Most of the industrial estates were established and built between 1989 and 1990—except for Deltamas (Greenland International Industrial Centre), which was built in the early 2000s. Several of these companies were jointly created with foreign investors. For instance, Bekasi Fajar Industrial Estate and Marubeni Corporation from Japan established a joint company to build the MM2100 industrial estate. East Jakarta Industrial Park is also co-owned by a Japanese investor, Sumitomo Corporation. Another Japanese investor, Sojitz Corporation, made a strategic alliance with Sinarmas Land, a

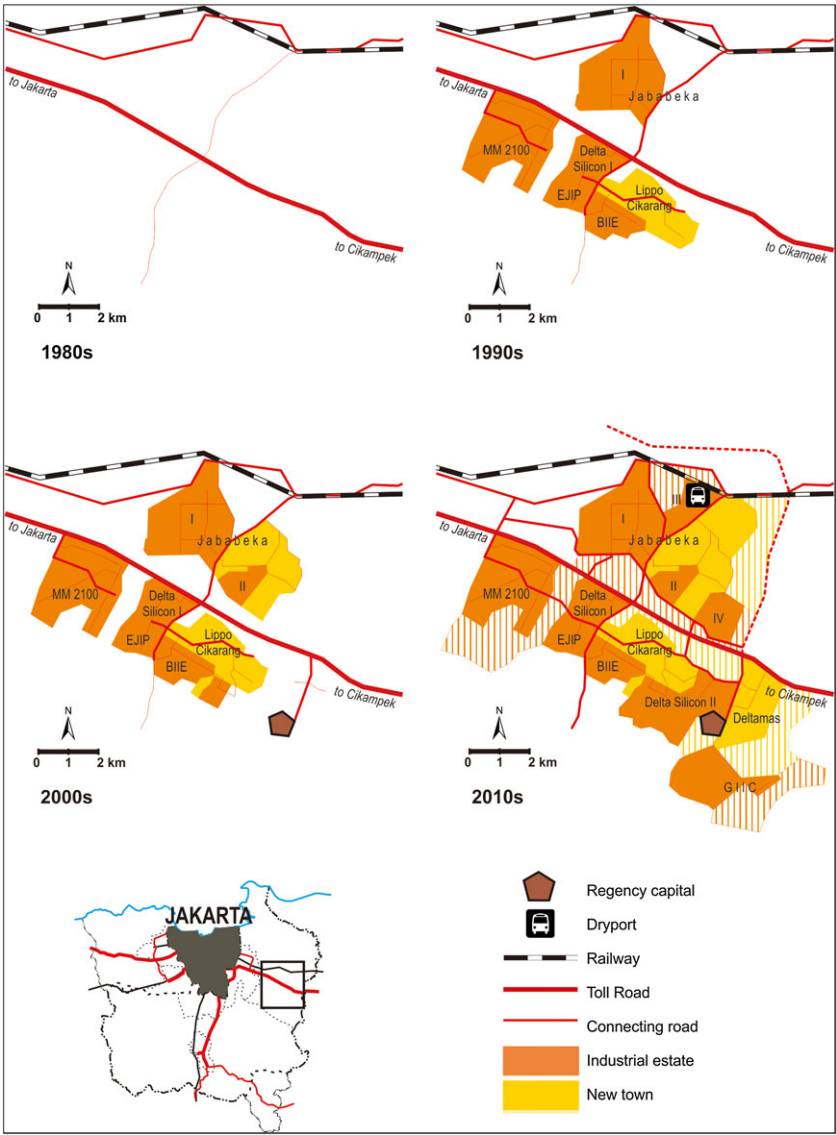


Figure 3. Growth of industrial estates and new towns in Cikarang from 1980s to 2010s.
Source: Author’s analysis.

Table 1. The industrial estates in Cikarang.

Industrial estate	Granted area size (Ha)	Country of origin (company)	Year of operation/development
Bekasi Fajar Industrial Estate	2100	Indonesia (Bekasi Fajar Industrial Estate)	1989
Deltamas (Greenland International Industrial Centre)	3000	Japan (Sojitz Corporation) and Indonesia (Sinarmas Group)	Early 2000s
East Jakarta Industrial Park	320	Japan (Sumitomo Corporation)	1989
Hyundai Inti Development	200	South Korea (Hyundai Corporation) and Indonesia (Bekasi International Industrial Estate)	1989
Jababeka	5600	Indonesia (Jababeka Group)	1989
Lippo (Delta Silicon)	2500	Indonesia (Lippo Group)	1989
MM2100	1200	Japan (Marubeni Corporation) and Indonesia (Bekasi Fajar Industrial Estate)	1989

Sources: Lippo Cikarang (2009); Hyundai Inti Development (2010); Jababeka (2010); MM2100 Industrial Town (2010); East Jakarta Industrial Park (2014).

subsidiary company of Indonesia's largest developer (Sinarmas Group) to initiate the industrial estate project of Greenland International Industrial Centre located in the Deltamas new town. Meanwhile, Hyundai Inti Development was the result of cooperation between Bekasi International Industrial Estate and Hyundai Corporation, South Korea. Other industrial estates were built by Indonesia's largest conglomerates and property developers, Lippo Group (Delta Silicon) and Jababeka.

Jababeka is the largest industrial estate company in Cikarang and is probably the only private industrial estate in Cikarang that reflects a considerable level of diversity. Their tenants largely include small and medium-sized companies from more than 30 countries (Jababeka, 2011). In comparison, the other six industrial estates attract only groups of large-scale multinational companies from a select few countries (Table 2). The Japanese co-owned industrial estates, MM2100 and East Jakarta Industrial Park, are traditionally dominated by Japanese multinational companies. Likewise, Bekasi International Industrial Estate (Hyundai) from the beginning envisioned to support Korean high-tech companies. Clearly, globally linked industrial estate development in Cikarang tends to stimulate segregation by nationalities, probably because of the cooperation strategies commonly used by the East Asian multinational companies to promote vertical integration in their global production networks (Hatch & Yamamura, 1996).

Initially, most developers focused their business operation on purely industrial estate development and management. However, the massive FDI inflows, which peaked in the

Table 2. Investors by country of origin in four industrial estates in Cikarang (2010).

Origin of investor	MM2100	Bekasi International Industrial Estate	East Jakarta Industrial Park	Delta Silicon
Domestic	35	75	4	20
Foreign	141	37	77	93
Japan	121	6	73	13
Korea	6	24	0	9
Taiwan	1	3	1	26
Other	13	4	3	45
Total	176	112	81	113

Source: Analysed from Institut Teknologi Bandung (2012).

mid-1990s, have resulted in a rapidly growing number of migrant workers and population. Within 20 years, the population of industrial districts in Cikarang grew rapidly from less than 5000 in 1989 to almost more than 1.2 million in 2009 (Hudalah & Firman, 2012). The associated population growth generated huge multiplier effects related to the fulfillment of daily economic and social needs. Given these new opportunities, Jababeka gradually diversified its business orientation. While the first project (Jababeka I) was focused on developing and managing industrial land, the second one (Jababeka II), which was started in the early 2000s, attempted to capture a growing demand for housing and urban facilities.

Several external factors in the 1980s to 1990s such as the deregulation packages in banking and financial sectors, high economic growth and a rising number of middle-income households have encouraged big developers to shift their business orientation to creating new exclusive living environments in the periurban areas (Winarso & Firman, 2002). For instance, Lippo Cikarang (2009) envisioned *kota mandiri* or 'self-sufficient towns', by providing a 'place to work, to live, and to play' (Table 3). In general, the industrial estates in Cikarang successfully built industrial, commercial and residential areas along with premium education, healthcare and leisure facilities such as private universities and schools, international standard hospitals, hotels, shopping malls, recreational parks and entertainment centres (Hudalah & Firman, 2012).

The private living environment tends to transform into a property machine that divides the growing number of inhabitants into fragments based on social classes (Pérez, 2002; Merrifield, 2013). In periurban Jakarta, exclusive middle-upper income housing construction has facilitated the formation of gated communities and reinforced social segregation in and surrounding the new towns (Leisch, 2002; Firman, 2004). The integration of locality into the global economy may further increase these inherent inequalities and divisions within the local economies (Gugler, 2004; Palit, 2009). In

Table 3. Visions and missions of industrial estates and new towns in Cikarang.

Industrial estate	Vision	Mission
Jababeka	'To become the most respected and environmentally friendly township developer'	'Reliable, professional and always striving to exceed expectation'
Lippo Cikarang	'Place to work, to live, and play'	'To become a leading industrial, commercial, and residential-based urban developer in Indonesia, through urban infrastructure, public facility, and managerial investment ...'
Bekasi International Industrial Estate (MM2100)	'To be Indonesia's leading industrial provider of superior business environment and services for manufacturing companies and related operations'	
East Jakarta Industrial Park	'Global Corporate Making great Contributions to Society'	'To become a leading industrial park in Asia ...'
Deltamas	'A Better Living'	'To create a pleasant, harmonious and sustainable living environment for its residents'
Bekasi Fajar Industrial Estate (Hyundai)	'To create a high-technology and comprehensive industrial park to support Korean companies'	

Sources: Lippo Cikarang (2009); Hyundai Inti Development (2010); Jababeka (2010); MM2100 Industrial Town (2010); East Jakarta Industrial Park (2014).

Cikarang, the emergence of FDI-induced private industrial estates has severely widened income gaps in Bekasi District (Hudalah & Firman, 2012).

The rise of local government: Indonesia's transition to democratic decentralization and the formation of International Zone

The rapid settlement transition in Cikarang in the past three decades can be partly explained by interlinking the shifting global factors with the dynamics of domestic institutional frameworks (Table 4). The 1997 Asian financial crisis reminded the manufacturing industries in Cikarang that the real competition was no longer with neighbouring industrial estates but with new emerging periurban areas in the metropolitan region and other industrializing Asian countries. This tightening regional and global competition forced the manufacturing production system in Cikarang to become more efficient.

Furthermore, the Asian financial crisis triggered massive student demonstrations in Indonesia and turned into a political chaos, which was signified by the stepping down of President Suharto and, thus, the end of the 33 years of authoritarian and centralized rule under the New Order Regime. Following this critical juncture in 1998, the country transformed into the largest democracy in Southeast Asia (Kimura, 2010). A series of radical democratization policy measures was drafted, marking the beginning of *era reformasi* (reform era).

Amid the risk of national disintegration, the Indonesian government enacted the first law on regional administrations in 1999 after the fall of the New Order regime. With the enactment of Law 22 of 1999 on Regional Administrations and Law 25 of 1999 on Inter-Government Financial Balance, Indonesia experimented with a radical shift from the centralized, authoritarian regime toward a highly decentralized administrative system. This earliest version of *otonomi daerah* or Indonesia's decentralization policy was considered very ambitious. The new laws entailed the devolution of a wide range of political powers in most policy areas at the lowest administration level or local government, including land use planning and infrastructure development. In fact, the local government could execute some of these policies without close consultation with central and provincial governmental administrations and neighbouring local governments. Hence, the *Dewan Perwakilan Rakyat Daerah* (DPRD) or the elected local council had more power in making laws and holding the head of local government accountable. In particular, Law 25 of 1999 also gave local governments the power to generate

Table 4. Periurban settlement transition in Cikarang.

Transition element	Industrial estate/town (1980s–1990s)	The international economic zone (2000s–2010s)
Global factor	Global FDI expansion	Asian and global financial crises
Domestic institutional settings	Deregulation and debureaucratization	Decentralization and democratization
Mode of governance	Shadow government	Network governance
Instrument	Privatization	Public-private partnership
Scale of state intervention	National-regional	Local-regional
Scale of competition	Local	Regional-global
Key political objective	Growth	Sustainability
Planning policy issue	Accessibility	Connectivity
Spatial manifestation	Enclave, splintering urbanism	Networked-enclave, bounded zone

Source: Author's analysis.

local revenues through taxes, charges and fees from local businesses and its citizens (Kimura, 2010).

These pervasive powers as a result of the decentralization euphoria that occurred throughout the country drove local governments to act as *raja kecil* (little kings) with their own autonomous territories and citizens (Firman, 2009). As Hadiz (2004:705) comments, 'rather than breaking up centralized state intervention and promoting investment, decentralization has so far opened the door to a host of uncertainties which frighten investors'. In addition to the ones promulgated by the central government, every autonomous region such as Bekasi Regency designed their own local regulations and policies with respect to investment and land and property development. These additional regulations complicated procedures and, thus, increased uncertainties and transaction costs for industrial investment and production in Cikarang.

For these reasons, in 2006 Jababeka, being the largest industrial estate developer in Cikarang, initiated the establishment of an association called Zona Internasional (ZONI) or International (economic) Zone. ZONI is a network-based institutional arrangement, mainly focused on creating a good environment for sustainable investment and development acceleration in Cikarang. The initiative was supported by the central government (the Ministry of Public Works), West Java Provincial Government, Bekasi Regency Government, the state-owned highway corporation (PT Jasa Marga) and the seven industrial estates in Cikarang. Zoni can be seen as a 'worlding practice' (Roy & Ong, 2011), envisioning Cikarang as an excellent and competitive industrial district in Asia (Jababeka, 2007).

ZONI conducted a feasibility study to propose Cikarang as a special economic zone (SEZ). To some extent, this proposal attempted to replicate successful elements of *kaifagu*, the Chinese development zone (Hsing, 2010). The SEZ would focus on debureaucratization and investment service integration to improve the business climate in Cikarang. Furthermore, ZONI also worked to improve the bargaining position of the industrial estates in local, regional and national policymaking. Under ZONI, the industrial estates attempted to enhance communication, cooperation and partnerships with the governments. In the following section, cooperation among the industrial estates over transport infrastructure development illustrates the challenges and prospects for ZONI.

Dynamics of periurban governance

Rise of shadow government

Rimmer and Dick (2009) note that periurban governments in Southeast Asia have limited resources in the provision of infrastructure. Most roads in Cikarang can only accommodate local traffic such as cars and motorcycles and cannot serve heavy vehicles such as container trucks. As a result of insufficient financial resources by the local government, the roads in Cikarang are partly maintained and taken care of by the government of Bekasi Regency. To make things worse, the decentralization policy has not significantly improved the fiscal condition of Bekasi Regency in general and Cikarang in particular. Despite the fact that the area has long become the prime location for factories, the manufacturing offices are still located in Jakarta. Consequently, Jakarta still control the largest share of companies' income and taxes, while Bekasi Regency could only generate limited revenues from land, building, and property taxes and permit fees.

In fact, although Cikarang has transformed into a national manufacturing powerhouse since mid-1990s, Government Regulation 26/2008 on the National Spatial Plan continued to treat Cikarang and other periurban areas in Bekasi District merely as a *kawasan penyangga* (dormitory or satellite town) with the function to support the residential expansion of the Special Capital Province of Jakarta (DKI Jakarta). A top official of Bekasi Regency Regional Development Planning Agency explained as follows:

The central government [still] thinks that Bekasi is a dormitory [town]. Meanwhile, [if you] try [to travel] from Jakarta to Cikarang in the morning, in the evening, during the weekdays, [the traffic will always busy and] only empty in the weekend. Cikarang is different from other [periurban] areas surrounding Jakarta. You cannot treat Cikarang like a satellite city of Jakarta. Instead, we can say that the [national] economic base is in Cikarang. However, the central government has not realized that (pers. comm., Cikarang, 20 March 2013).

In addition, the head of the Spatial Planning and Human Settlement Department of Bekasi Regency emphasized that Cikarang ‘has already been called an international economic zone yet [is] governed by a “rural” organization’ (pers. comm., Cikarang, 26 March 2013). The local government is unprepared for dealing with globally linked rapid urbanization, because of the Indonesia’s government structure. Indonesia’s constitution divides the governmental administration into three levels: central, provincial and local (People’s Consultative Assembly, 2002). Local governments are divided into *kota* (municipalities) and *kabupaten* (regencies). While *kota* covers an urbanized area, *kabupaten* is historically occupied by predominantly rural areas. As a form of rural government, *kabupaten* does not have adequate institutional capacity to deal with globally linked rapid urbanization.

As also found in other Asian cities, the limited capability of local government in responding to rapid industrialization and urbanization in the periurban fringe often allows large private developers to play an active and exclusive role in the infrastructure planning and development processes (Rimmer & Dick, 2009; Sood, 2015). To support the industrial operation, developers in Cikarang built the local roads, clean water treatment plants, waste water treatment plants, communication network and public transportation networks inside and surrounding their project sites. Because of their capacity in taking over the government’s tasks in providing infrastructure networks, the developers thought that they acted as a shadow government:

It would be difficult for us, for example, to invite investors to build factories but there was no road. They surely wouldn’t come. We, as developers, [therefore] acted as *pemerintah bayangan* [shadow government] (private developer, pers. comm., Cikarang, 6 March 2012).

In fact, the actual power of this shadow government might stretch beyond that featured in US edge cities (Garreau, 1991). A shadow government, which can take the form of corporate township in South Asian conception (Jenkins *et al.*, 2015; Sood, 2015), can be viewed as a nonelected or nonrepresentative organization that governs exclusive or gated space and operates outside the existing structures of local and regional governments. Such an organization is formally registered and its operation is regulated; thus, it can be distinguished from illegal organizations and criminal practices (see for example Wilson, 2009; Aspinall & Klinken, 2011). However, in practice a shadow government may operate in grey policy areas where the authority and legitimacy of the local government are not clearly specified and when local governments do not have financial, human and institutional capacity.

Garreau's conception of a shadow government restricts the active role of the private sector in creating and enforcing order at the local scale within their own project sites. Meanwhile, in Cikarang private developers extended their power into external realms, by directing the formulation of national and local legislation and by building parts of local and regional infrastructure networks located outside their project sites. The industrial estate developers guided the central and local governments in drafting laws, regulations and policies to further their own business interests. Being more experienced in managing urban growth, the developers also often pushed the regency government to adopt industrial estate regulations in the local spatial planning policy.

Furthermore, as also evident in the Philippines (Shatkin, 2008), large private developers in Cikarang have played an active role in realizing the global aspiration for regional infrastructure projects. The industrial estate developers in Cikarang have always had a common interest in shortening the distance to Jakarta, where the international seaport and airport for export-import processing are located. The four industrial estates improved their access to the intercity toll road, by proposing five new toll road interchanges only 3 to 6 km apart from each other and financing and building roads and bridges directly connected to the interchanges (Figure 4).

Toward network governance?

Although industrial estate developers have built an excellent infrastructure network within their own project sites, the infrastructure networks are disconnected from each other, leading to a fragmentation of the periurban space. The situation perfectly resembles what Graham and Marvin (2001) call the 'splintering urbanism'. To resolve the local infrastructure fragmentation, in 1993 the central government through the



Figure 4. A toll road interchange built by a private industrial developer in Cikarang.

Source: Photograph by author, 2013.

Industrial Estates Coordination Team initiated a handshake agreement with the provincial and local governments and the industrial estate developers in Cikarang. As part of this public-private partnership agreement, each industrial estate company was required to build connecting roads to their neighbouring project sites and local communities. Initially, most companies undermined this agreement. They argued that connecting roads would generate more traffic on their local roads, thus causing congestion and increasing the road maintenance costs. Such an externality would only benefit their neighbouring competitors who wish to get a free and faster access to the nearest interchanges that connect their project locations with Jakarta.

However, the lack of connectivity between the neighbouring industrial estates forced regional and local traffic to rely on the intercity toll road. After a decade, the situation created congestion and bottlenecks on the toll road and its interchanges. Several industrial estate managers in Cikarang started to worry about the impact of this poor connectivity on land value, production efficiency and the attractiveness of Cikarang as the national manufacturing centre. Furthermore, as the industrial activities grew and the production system complexity increased, the project sites could no longer be isolated from each other. Factories needed to better connect to their vendors that were increasingly dispersed across neighbouring project locations. Similarly, as the population grew, the industrial towns have increasingly specialized and neither of them could independently provide all facilities and services required by their own inhabitants.

Several industrial estate managers finally saw mutual interests and reciprocal benefits for improving infrastructure connectivity in Cikarang. They also realized that poor connectivity was increasingly becoming a common problem for major industrial estate developers in Cikarang. Hence, they attempted to initiate communication by building several communities and forums such as the President Executive Club (for professional and business leaders), Golf Country Club (for hobbies and leisure) and ICT forum (for engineers). They aimed to transform the embedded competition into partnership and cooperation.

So, in the beginning we cheated against each other, fought with each other. Each of us wanted to be the champion. We blocked the [connecting] road. 'I'm sorry. I don't want to connect [my road with yours]; I'm greater than you. I don't want to share my [drinking] water [with you]'. Nevertheless, at last, we realized that it was not efficient. Traffic jam was everywhere ... So the formula is, from now on, to make friends. I just realized why *silaturahmi* [to connect ties with friends] would expand our wealth. Evidently, [if we have] many friends, our business will be simpler. It did become simpler. Everything became cheaper ... Therefore, we shift toward networking. The current competition is between networks, not between entities ... If we don't have network, [we will be] abandoned, *wassalam* [goodbye] (industrial manager, pers. comm., Cikarang, 20 March 2013).

Using this networking strategy, in 2006 the industrial estate managers and the governments renewed the partnership agreement on the infrastructure development and improvement programmes to support the industrial activities in Cikarang. Following the new joint agreement (Ministry of Public Works, 2006), the central government focused on building new toll roads, flyovers and bridges, and the Indonesia Highway Corporation was given the authority to build new toll road lanes. The main roads were the responsibility of the provincial government, while the local roads were upgraded by the regency government. Meanwhile, most connecting roads and interchanges were constructed by the private industrial estate companies (Figure 5).



Figure 5. A connecting road built through public-private partnership in Cikarang.

Source: Photograph by author, 2013.

Conclusion

In the past three decades, mega property developments in the form of industrial estates, towns and regional infrastructure have spread across JMA's periurban areas. The relocation of FDI on the global scale, especially in manufacturing sectors, is a major driver of this urban expansion. Furthermore, the local spatial representation of the global economic factor is not independent but strategically interlinked with periurban politics. Shifts in domestic institutional settings, particularly Indonesia's rapid transition to democratic decentralization, have increased the significance and the complexity of periurban governance, which faces a key challenge of reconnecting the fragmented infrastructure networks. The governing model of periurban infrastructure is not static and state centred. Instead, the elements of governance, including the spatial scales, state-market relations, mechanisms and instruments, and policy objectives are highly diverse and dynamic.

At least two significant modes of governance emerge in the case study: shadow government and network governance. First, the private sector acted as the so-called shadow government, in the face of limited capability of local government. My analysis implies that even in the context of underdeveloped governmental institutions, the effectiveness of this private governance model is limited in fostering integrated and inclusive development and improving the competitiveness of periurban infrastructure networks.

Second, the building of network governance that cuts across different actors and scales could be crucial in rapidly periurbanizing metropolitan areas such as JMA. While the government highlights the authority and coercive power of the state, network governance consists of selected autonomous actors whose capacity in enabling collective actions lies in social coordination and control (Jones *et al.*, 1997). In the case of Cikarang, interactor networking strategies seem to be helpful in reconciling the state-market divides and resolving the global-local political contestation. The strategies may include seeking mutual interests, reciprocal benefits and common problems, and initiating

communication, cooperation and partnerships. However, because of a lack of trust, Zoni as a private-driven institution currently finds difficulties in encouraging quick actions and building long-term commitment of its members to promote sustainable urban development. Given the limited commitment of private developers, the institutional, financial and human capacities of local and regional governments need to be upgraded.

Acknowledgements

The author wishes to thank Nicholas Phelps, Dimitra Viantari, Ratih Dewayanti and Ishma F. Soepriadi for the fruitful discussions and their assistance during the fieldwork for this paper. However, the author solely responsible for the content of this article.

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